Integers, Primes & Powers — the building blocks of number

NAME _____ DATE _____ SCORE / EFFORT _____

Learning intentions — by the end of this worksheet I can:

- carry out all four operations with integers and follow the order of operations correctly
- write any whole number as a product of primes in index form
- use prime factorisation to find the HCF and LCM, and to recognise perfect squares and cubes
- solve unfamiliar problems by choosing the right number tool

Part A · Warm-up (diagnostic) ⏱ ~6 min

No calculator for the whole worksheet. Show working where it helps.

1. $\frac{3}{4} + \frac{2}{3} =$ _____
2. 15% of 240 = _____
3. $0.6 \times 0.07 =$ _____
4. $2\frac{1}{2} \div \frac{5}{6} =$ _____
5. 48×25 (find a clever mental method) = _____
6. Write 3.14159 correct to two decimal places: _____

Part B · Learn ⏱ ~8 min

1. Integer rules

Operation	Rule	Example
Adding a negative	same as subtracting	$5 + (-8) = 5 - 8 = -3$
Subtracting a negative	same as adding	$-4 - (-9) = -4 + 9 = 5$
× or ÷, **same** signs	positive	$(-6) \times (-7) = 42$
× or ÷, **different** signs	negative	$-56 \div 8 = -7$

Watch out: $(-3)^2 = (-3) \times (-3) = 9$, but $-3^2 = -(3 \times 3) = -9$. The index applies only to what it is touching.

2. Order of operations

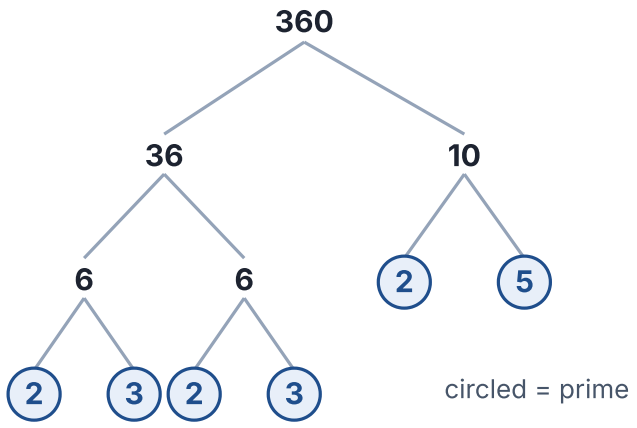
Brackets → **I**ndices (powers & roots) → **D**ivision & **M**ultiplication (left to right) → **A**ddition & **S**ubtraction (left to right).

Worked example 1: $-3 \times (-4) - 18 \div (-6)$
$= 12 - (-3)$ *(do × and ÷ first)*
$= 12 + 3 = 15$

3. Prime factorisation and index form

A **prime** has exactly two factors: 1 and itself (2, 3, 5, 7, 11, 13, ...). **1 is not prime.**
Every whole number greater than 1 can be written as a product of primes in exactly one way (the *Fundamental Theorem of Arithmetic*).

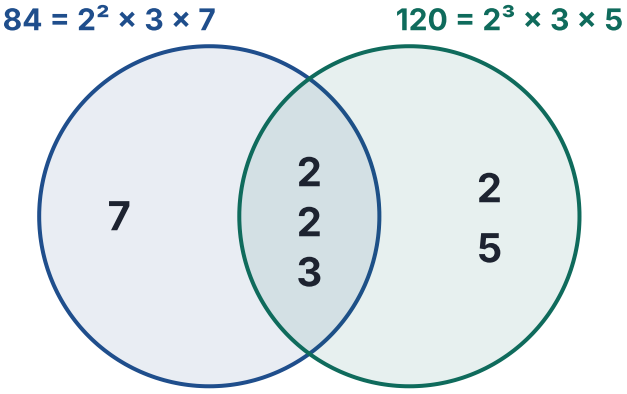
Worked example 2: $360 = 36 \times 10 = (2^2 \times 3^2) \times (2 \times 5) = 2^3 \times 3^2 \times 5$



4. HCF and LCM from prime factors

- **HCF** (highest common factor): take the primes the numbers **share**, each to the **lowest** power.
- **LCM** (lowest common multiple): take **every** prime that appears, each to the **highest** power.

Worked example 3: $84 = 2^2 \times 3 \times 7$ and $120 = 2^3 \times 3 \times 5$
HCF $= 2^2 \times 3 = 12$ LCM $= 2^3 \times 3 \times 5 \times 7 = 840$
Check: HCF × LCM $= 12 \times 840 = 10\,080 = 84 \times 120$ ✔ (true for any two numbers — can you see why?)



overlap → HCF = 2 × 2 × 3 = 12 all numbers → LCM = 2·2·3·7·2·5 = 840

5. Squares, cubes and roots

A number is a **perfect square** when every index in its prime factorisation is even, and a **perfect cube** when every index is a multiple of 3.
$\sqrt{2^4 \times 3^2} = 2^2 \times 3 = 12$ (halve the indices) $\sqrt[3]{2^6 \times 5^3} = 2^2 \times 5 = 20$ (divide the indices by 3)

Part C · Practice ⏱ ~25 min

C1 ⭐ Fluency

1. Evaluate.

a) $-17 + 9$	b) $-6 - (-14)$	c) -8×7
d) $-72 \div (-9)$	e) $(-3)^3$	f) -3^2
g) $5 - 3 \times (-4)$	h) $(-2 - 6) \times (-5) + 4$	i) $-24 \div (4 - 10) - (-3)^2$
j) $[18 - 3 \times (2 - 7)] \div (-11)$		

WORKING

2. Write each number as a product of primes in index form.

a. 72 _____ b) 150 _____ c) 196 _____
b. 1001 _____ e) 2024 _____ f) 3600 _____

WORKING

3. Evaluate.

a. 2^5 _____ b) 3^4 _____ c) 10^6 _____ d) $\sqrt{169}$ _____
b. $\sqrt{2^4 \times 3^2}$ _____ f) $\sqrt[3]{343}$ _____ g) $4^3 - 3^4$ _____

C2 ⭐⭐ Understanding

4. Use prime factorisation to find the HCF and LCM.

Numbers	Prime factorisations	HCF	LCM
36 and 60			
45 and 98			
24, 40 and 90			

WORKING

What is special about 45 and 98?

5. Insert brackets so each statement is true.

a. $6 + 4 \times 3 - 1 = 29$
b. $6 + 4 \times 3 - 1 = 20$
c. $6 + 4 \times 3 - 1 = 14$
d. $-2 \times 5 - 8 \div 2 = 3$

6. True or false? Justify each answer with a reason or a counter-example.

a. The product of two negative integers is always greater than their sum.

b. Every prime number is odd.

c. $(-1)^n = -1$ for every odd whole number n.

d. The LCM of two numbers is always their product.

7. Find the error. Sam's working:

$20 - 12 \div 4 \times (-3) = 8 \div 4 \times (-3) = 2 \times (-3) = -6$

Explain Sam's mistake and give the correct answer.

8. At an Australian research station in Antarctica the temperature at 6 am is $-27\,^{\circ}\text{C}$. It rises by $3\,^{\circ}\text{C}$ every hour until 2 pm, then falls by $5\,^{\circ}\text{C}$ every hour until 9 pm.

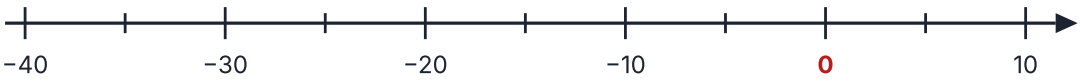
a) What is the temperature at 2 pm? _____

b) What is the temperature at 9 pm? _____

c) What is the difference between the highest and lowest temperatures recorded that day (6 am – 9 pm)? _____

WORKING

°C (use this number line to track the temperature)



C3 ⭐⭐⭐ Reasoning

9. Use prime factorisation (not a calculator!) to decide whether each number is a perfect square, a perfect cube, both, or neither. Give the root where it exists.

a. 1764 _____ b) 5832 _____ c) 2700

WORKING

10. For 2700 in Q9:

a) What is the smallest whole number you could multiply it by to make a perfect square? _____

b) ...to make a perfect cube? _____

WORKING

11. Find the smallest positive integer n such that $168n$ is a perfect square. What is $\sqrt{168n}$?

WORKING

C4 🔗 Connections — this week's tools meet earlier topics

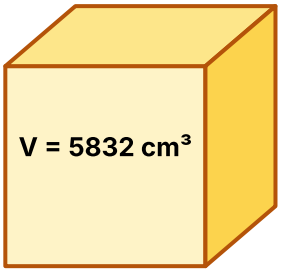
Prime factors, HCF and LCM aren't just for number puzzles. Use them inside topics you already know (area, volume, fractions, substitution).

A



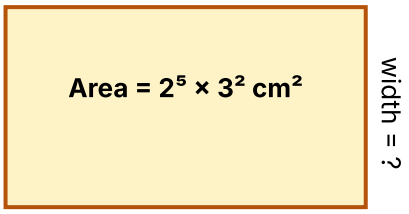
side = ?

B



edge = ?

C



length = 2^4 cm

X1. Shape A (area & perimeter). The square courtyard has an area of 1764 m². Use its prime factorisation (Q9) to find the side length without a calculator, then find its perimeter.

WORKING

X2. Shape B (volume & surface area). The cube-shaped box has a volume of 5832 cm³. Find the length of each edge, then the box's total surface area.

WORKING

X3. Shape C (area & index form). The rectangle has area $2^5 \times 3^2$ cm² and length 2^4 cm. Find the width in index form *and* as an ordinary number. Then find the perimeter.

WORKING

X4. Fractions. Simplify $\frac{84}{120}$ in **one step** using the HCF from Worked example 3.

WORKING

X5. Fractions. Use the LCM as the lowest common denominator.

a) $\frac{5}{12} + \frac{7}{18}$ _____ b) $\frac{1}{24} - \frac{1}{40}$ _____

WORKING

X6. Substitution (algebra preview). If $a = -3$ and $b = 4$, evaluate:

a) $a^2 - 2ab$ _____ b) $(a - b)^3$ _____ c) $\frac{b^2 - a}{a + b}$ _____

WORKING

Part D · Problem Solving ⏱ ~12 min

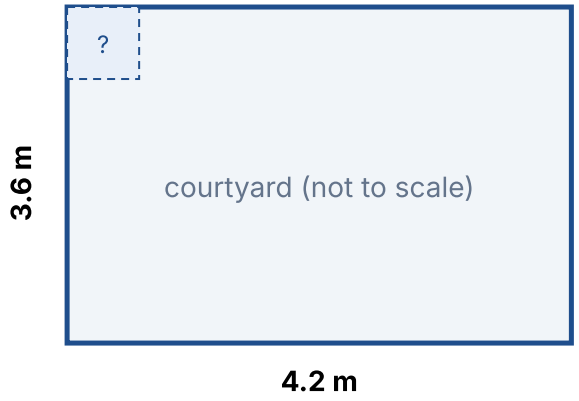
12. Buses. Suppose two buses leave Elizabeth Quay Bus Station together at 7:00 am. Route A departs every 12 minutes and Route B every 18 minutes.

a) When do they next leave together? _____

b) How many times do they leave together from 7:00 am to 10:00 am (inclusive)? _____

WORKING

13. Tiling. A rectangular courtyard measures 4.2 m by 3.6 m. It is to be covered with identical square tiles with **no cutting**. What is the largest tile size possible, and how many tiles are needed?



WORKING

14. Packing. A school fundraiser has 84 muesli bars, 126 juice boxes and 210 apples. They are packed into identical bags with nothing left over. What is the greatest number of bags possible, and what goes in each bag?

WORKING

15. Quiz scoring. In a 25-question quiz, a correct answer scores +4 and an incorrect answer scores −2.
a) Amira answered every question and scored 46. How many did she get correct? _____

WORKING

b. Leaving a question blank scores 0. Is it possible for anyone to score exactly 47? Explain using odd/even reasoning.

Part E · Challenge ⏱ extra

16. Counting factors. $72 = 2^3 \times 3^2$ has 12 factors.

a) List them all to check.

b. Explain why the number of factors of $2^a \times 3^b$ is $(a+1)(b+1)$.

c. Find the **smallest** whole number with exactly 12 factors.

WORKING

17. What is the last digit of 7^{2026}? Explain the pattern you used.

18. Explain why the product of any three consecutive integers is always divisible by 6.

Self-check

I can…	😀	🙂	😕
use integer rules and the order of operations without errors			
write a number as a product of primes in index form			
find HCF and LCM from prime factorisations			
recognise perfect squares/cubes from prime factorisations			
choose HCF or LCM in a worded problem			

⚡ Fast-finisher bank ⏱ ~35 min

Work through in order. ⭐⭐⭐⭐ = competition / Year 9+ level. No calculator.

F1 ⭐⭐ Evaluate.

a) $(-2)^5 - (-3)^3$ _____ b) $[(-6)^2 - 4 \times (-3) \times 2] \div (-2)^2$ _____

c) $-2^4 + (-2)^4$ _____ d) $3 - 4 \times [5 - (-2)^3] \div 13$ _____

WORKING

F2 ⭐⭐ Write 9450 and 13 860 as products of primes, then find their HCF and LCM. Check your answer with HCF × LCM.

WORKING

F3 ⭐⭐⭐ Two numbers have HCF 12 and LCM 360. One of them is 72. Find the other, and check it.

WORKING

F4 ⭐⭐⭐ Find **all** pairs of positive integers whose HCF is 6 and whose LCM is 72.

WORKING

F5 ⭐⭐⭐ Gears. A gear with 24 teeth meshes with a gear with 36 teeth. A tooth on each gear is marked where they touch. How many full turns must each gear make before the two marked teeth touch again?

WORKING

F6 ⭐⭐⭐ What is the smallest perfect square that is divisible by 2, 3, 4, 5 and 6?

WORKING

F7 ⭐⭐⭐ $25! = 25 \times 24 \times 23 \times \ldots \times 2 \times 1$. Without calculating it, how many zeros are on the end of $25!$? Explain using prime factors. *(Hint: each zero needs one 2 and one 5.)*

F8 ⭐⭐⭐ If $2^a \times 3^b = 6^4 \times 12$, find a and b.

WORKING

F9 ⭐⭐⭐⭐ **Sum of factors.** The sum of all factors of $360 = 2^3 \times 3^2 \times 5$ can be found by calculating $(1 + 2 + 4 + 8)(1 + 3 + 9)(1 + 5)$.

a) Calculate it.

WORKING

b. Explain why this works. *(Think about what you get when you expand the brackets.)*

c. Use the same method to find the sum of the factors of 72.

WORKING

F10 ⭐⭐⭐⭐ **The locker problem.** A corridor has 100 closed lockers numbered 1–100. Student 1 opens every locker. Student 2 changes every 2nd locker (closes it if open, opens it if closed). Student 3 changes every 3rd locker, and so on, up to Student 100.
a) Which lockers are open at the end? b) Explain why, using what you know about factors.

F11 ⭐⭐⭐⭐ **Investigation — an unsolved problem.** In 1742 Christian Goldbach suggested that every even number greater than 2 is the sum of two primes. Nobody has ever proved it.
a) Check it for every even number from 4 to 30.

b. In how many different ways can 50 be written as the sum of two primes? (3 + 47 and 47 + 3 count as the same way.)

c. Why does the conjecture say *even* numbers? What goes wrong for odd numbers like 27?